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EXECUTIVE SUMMARY OF THE THESIS

A Novel Digital Health Competence Framework for Patients: Model Validation and Application in Italy

TESI MAGISTRALE IN MANAGEMENT ENGINEERING – INGEGNERIA GESTIONALE

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1. Introduction

With the advent of the internet and its network effect, public access to medical information increased, gradually reducing informational asymmetries between providers and patients. Advancements in healthcare technology and its digitisation, can improve the quality of its services, but when digital health competence is uneven across the population, these advancements risk reinforcing or widening existing inequalities. One of the key insights from prior studies of implementing digital health interventions was the fact that access to the potential benefits of these tools was uneven. It might be beneficial for some, but at the same time can be disadvantageous for certain groups. People from low-income backgrounds, different physical and cognitive aptitudes, lower educational levels, older age, and residing in rural areas, frequently report factors which show unequal outcomes of the digital health interventions applied. These findings indicate that social and structural vulnerabilities shape who can benefit from digital health interventions. This phenomenon is an extension of the already present digital divide into the domain of healthcare systems. Individuals with low digital health competence are constrained in their ability to locate, understand and use online health information. Conversely people with higher digital health competence not only access this information better but are able to use them more effectively[1]. Taken together, this indicates that uneven distribution of digital health competence may reinforce or widen the already existing social and health inequalities. Hence a robust, reliable and updated framework for the assessment of Digital Health Competence is needed.

1.1 Digital Health

The most prevalent definitions of Digital Health are defined by the World Health Organisation (WHO) as, "the use of information and communications technology in support of health and health-related fields.
Telemedicine, on the other hand is defined as, the provision of healthcare services, where distance is a significant factor, by all healthcare professionals utilizing information and

communication technologies, with the objective of enhancing the health of individuals and communities[2]

1.2 Digital Health Competence Framework

Digital Heath Competence is one of the key domains of Digital Health and its implementation. It is described as the ability of individuals, teams and organizations to access, comprehend, evaluate, generate, communicate, and securely utilize digital health information, devices, and services to enhance health, provide care, and optimize healthcare system performance. A Digital Health Competence Framework is an assessment tool which incorporates several methods to examine and determine the level of digital health competence of an individual or an organisation in various contexts. The methods included may range from diverse statistical tests to simple questionnaires, depending on the history, geography and moment in time that the competence is being assessed[3].

The objective of this thesis is

(1) to create a novel digital health competence framework based on the previous literature in this domain,
(2) to statistically validate the framework by identifying key latent behaviours and the factors that influence them, and
(3) assess the current status of digital health competence in the sample Italian population, to provide actionable policy recommendations.

2. Theoretical Background

Over the years various frameworks have been developed to assess the level of digital health competence of various groups of individuals. Created by the European Commission termed as the DigComp 2.2 - Digital Competence for Citizens, serves as a reference model that explains a cohesive understanding of digital competence and offers a standardized vocabulary for its expression and application in policy, education, and evaluation. DigComp 2.2 is structured into five dimensions: (1) domains of competence; (2) identifiers and titles of competence; (3) levels of proficiency; (4) illustrations of knowledge, skills, and attitudes; and (5) applications enabling flexible utilization across several contexts (e.g., healthcare)[3]. Norman and Skinner (2006) developed and psychometrically assessed the 8-item eHealth Literacy Scale (eHEALS) to evaluate consumers' perceived competence in locating, assessing, and utilizing online health information. The study offered an effective and dependable single-factor assessment of perceived eHealth literacy and suggested validity across varied demographics and in relation to behavioral outcomes[4]. Longhini et al. investigated the methods employed in evaluating healthcare professionals' digital health competence and the specific constructs measured by those instruments. They reviewed work from 1999 to 2021, incorporating predominantly 26 cross-sectional research across various occupations and contexts. They found four recurring areas of interest from hundreds of these items, namely, self-assessed capabilities, attitudes and psychological aspects regarding technology, reported utilization of digital tools, and knowledge of digital technologies. The authors concluded that, although eHEALS provided a dependable assessment of perceived literacy, it was originally intended for consumers and so necessitated modification and new validation for clinical populations and settings. Vicente et al., argued that eHealth competence exhibited significant variation between EU Member States and demographic groups, necessitating that EU-level strategies be customized to local contexts[5]. It was inferred from the analysis of the literature, that the underlying composition of the overall digital health competence is evolving and encompassing newer dimensions as the technologies evolve. There is an urgent need to develop newer frameworks that do not solely rely on psychometric assessments, but helps in understanding the evolution of the competence, that can be further used for policy design and implementation.

3. Framework Design and Implementation

We designed a Novel Digital Competence Framework, tailored to the context of the Italian

population, to understand these key latent behaviours, the evolution and the factors that influence them. The framework was operationalised using a custom-made questionnaire. The questionnaire also helps in understanding certain areas which were found to be overlooked in the literature review. The questionnaire was designed to accurately reflect the current status of digital competence in the research environment, rather than an idealized or solely aspirational version. The questionnaire was created in collaboration with Digital Health Observatory of Politecnico Di Milano. The questions included in the questionnaire resulted in both binary responses and likert based responses. This questionnaire was in Italian and a sample population of 1000 people were chosen at random from different sociodemographic backgrounds. The survey was conducted using telephonic interviews, where the interviewees were asked these questions and then their responses recorded and organised.

3.1 Data Analysis and Validation

The dataset in its totality belonged to the category of unsupervised dataset. Unsupervised data are datasets which have only input data and these input data doesn't explicitly relate to a single factor or to each other. We wanted to understand the underlying latent behaviours which resulted in the responses recorded through the questionnaire and what kind of factors were influencing those behaviours. Thus to look beyond simple behaviours, we needed to adopt a two-part statistical test. For validating the underlying behaviours we adopted a Partial Least Square Structural Equation Modelling (PLS-SEM) and for the factors influencing the behaviours a Multiple Linear Regression with a restricted cubic spline model based on the results derived from the PLS-SEM. Based on the research of the Digital Health Observatory of Politecnico Di Milano, the general behaviours that were observed influencing the sample population were:

1. **Digital Literacy (DLIT)**: Technical competence associated with the operational capabilities relating to technologies utilized by citizens-patients to access digital services (personal computers, smartphones etc).
2. **Health Literacy (HLIT)**: The competence required to seek, evaluate, and comprehend pertinent information for making informed health decisions.
3. **Digital Soft Skills (DSS)**: Competence for effective communication with professionals or patients using digital platforms.
4. **eHealth Skill (EHS)**: Competence to utilize digital solutions in the healthcare sector responsibly, independently, and well.

For the PLS SEM the above general behaviours were regarded as the Latent constructs. To maintain the balance of formative and composite constructs, Digital Literacy (DLIT) and Health Literacy (HLIT) were regarded as reflective constructs and the later two as formative/composite constructs. The construct scores generated and extracted through PLS-SEM were expressed in z-scores. To improve interpretability while preserving statistical validity, the latent construct scores were rescaled to a 0–100 index using a min–max transformation, equation 1.

$$X_i^{index} = \left(\frac{X_i - \min(X)}{\max(X) - \min(X)}\right) * 100 \qquad (1)$$

This transformation ensures that the lowest observed value corresponds to 0 and the highest to 100, leaving all intermediate values distributed proportionally between these extremes.

In the second stage of the analysis, the rescaled construct scores were employed as dependent variables in a regression model to examine the extent to which socio-demographic factors influence digital and health-related competences[6]. One important thing to note here is that the influence of the reflective constructs (Digital Literacy & Health Literacy) over the formative constructs (Digital Soft Skills & eHealth Skill) was already established by the inner structural model of the PLS SEM with the percentages of variances for each accounted for. Consequently, this creates the situation where, any factor that influences the reflective constructs will inherently affect the formative constructs. Keeping this in mind we perform regression analysis only on the reflective

constructs i.e. Digital Literacy and Health Literacy. Initially, several of the predictors were not distributed evenly across the dataset. Incorporating them in this condition would've resulted in faulty and biased outcomes. To amend this, predictors below 5% of the total distribution were combined. Additionally, it was decided to factor "Age" as a non-linear regressor, through a constrained (natural) cubic spline with three degrees of freedom, executed as ns(AGE, df = 3) in R. This was done because, usually, as has been observed in several studies related to skill and age, the relationship of age with any type of dependent variable doesn't necessarily follow linearity.

The resulting multiple regression for $\boldsymbol{Y_i}$ = Digital Literacy & Health Literacy, was as in equation 2:

$$\boldsymbol{Y_i = \beta_0 + \beta_{age,1}\, s_1(AGE_i) + \beta_{age,2}\, s_2(AGE_i) + \beta_{age,3}\, s_3(AGE_i) + \beta_2\,(Sex = 2) + \beta_3\,(Amplitude) + \beta_4\,(Employment\ Status) + \beta_5(Highest\ Qualification) + \beta_6\,(Profession) + \beta_7\,(D1_1\ ) + \beta_8(Region) + \varepsilon} \quad (2)$$

4. Results

The measurement (outer) and structural models(inner) were assessed using Partial Least Squares Structural Equation Modelling (PLS-SEM) with 5,000 bootstrap resamples to evaluate indicator reliability, construct validity, and structural relationships.

4.1 Measurement Model

Reliability and validity statistics indicated that all constructs obtained the required thresholds for an optimum model. With composite reliability (ϱC) values exceeding 0.88 for every construct and Cronbach's alpha values ranging from 0.871 to 0.931, high internal consistency was confirmed. The average variance extracted (AVE) values showed convergent validity and above the suggested minimum of 0.50, ranging from 0.659 to 0.706. The dependability of the indicators was likewise satisfactory. Each outer loading, ranging from 0.697 to 0.922, were found to be statistically significant (p-value < 0.001). Strong item–construct relationships were indicated by the loadings with ranges of ≥ 0.79. In terms of the discriminant validity of the inner structural model, some components showed significant overlap rather than distinct separation, according to HTMT values. Discriminant validity issues are particularly raised by HLIT–EHS (HTMT =0.987) and DSS–EHS (HTMT =0.993).

4.2 Structural model

Digital Literacy (DLIT) accounted for 82.5% of the variance in Digital Soft Skills (DSS), and its effect on DSS was strong and significant (β = 0.908, t = 117.67, 95% CI, p-value < 0.001). A variance of 87.4% was explained by Health Literacy (HLIT), for eHealth Skills (EHS) (β = 0.935, t = 125.76, 95% CI, p-value < 0.001).

Construct	Cronbach's Alpha (α)	Composite Reliability (ρc)	Average Variance Extracted
Digital Literacy (DLIT)	0.931	0.931	0.693
Health Literacy (HLIT)	0.885	0.885	0.659
Digital Soft Skills (DSS)	0.917	0.932	0.697
eHealth Skills (EHS)	0.871	0.905	0.706

Table 1: Reliability And Convergent Validity of the Constructs

4.3 Interpretation of the PLS-SEM

The findings emphasize the value and necessity of existing competence in the development of new ones in the context of digital health competence. While EHS is mostly determined by HLIT, better DSS is linked to higher DLIT. 0.825 and 0.873, the high R2 values, imply a significant explanatory power. However, the exceptionally strong HTMT between some conceptions suggests that they may be empirically similar. All the values representing the reliability and the convergent validity of the PLS SEM are represented in table 1. The composition of the latent variables in regard

their own loadings and weights are illustrated in the figure 1.

4.4 Results of the regression

The DLIT model explains a modest share of variance ($R^2 = 0.176$, adjusted 0.145; $F(36, 963) = 5.709$, p-value $<2.2\times10^{-16}$; residual SD = 26), while the HLIT model has slightly lower explanatory power ($R^2 = 0.114$, adjusted 0.080; $F(36, 963) = 3.427$, $p = 1.14\times10^{-10}$; SD = 24.59), but both the regressions depict uneven distribution of the effect of the socio demographic factors. In both cases, age is a crucial non-linear predictor where literacy rises in early adulthood, plateaus, and declines notably after about 50. In terms of education, in both the cases, people with the level of education of Lower Secondary or below ($\beta = -5.07$, p-value = 0.066) show a strong negative effect. Occupational differences are modest and concentrated among general workers and housewives (DLIT $\beta = -8.35$, $p = 0.042$ and $\beta = -18.57$, p-value <0.001; HLIT $\beta = -6.92$, $p = 0.080$ and $\beta = -8.59$, $p = 0.093$). Regional differences are quite substantial given that Lazio, Puglia, Basilicata–Calabria and Sardinia show higher scores than the reference region by roughly 10 to16 points.

Finally, a pre-existing health condition is positively associated with both DLIT ($\beta = 3.88$, $p = 0.031$) and HLIT ($\beta = 6.31$, p-value <0.001).

5. Discussion and Policy Recommendations

The main aim of this thesis was to answer three questions. First, how do we create a novel Digital Competence Framework based on the existing literature, secondly, how do we uncover the latent behavioural and socio-demographic factors that affect the levels of digital health competence across the sample Italian population and finally, what actionable policy recommendations can be derived from these findings.For the first question, the framework that was created in collaboration with the Digital Health Observatory of Politecnico Di Milano accomplished the task by keeping in mind the problems which were highlighted in the literature review. Our framework was conceived as a practical tool to capture citizens' digital and eHealth competences and was structured so that its content mapped closely to the DigComp 2.2 framework of the European Union.

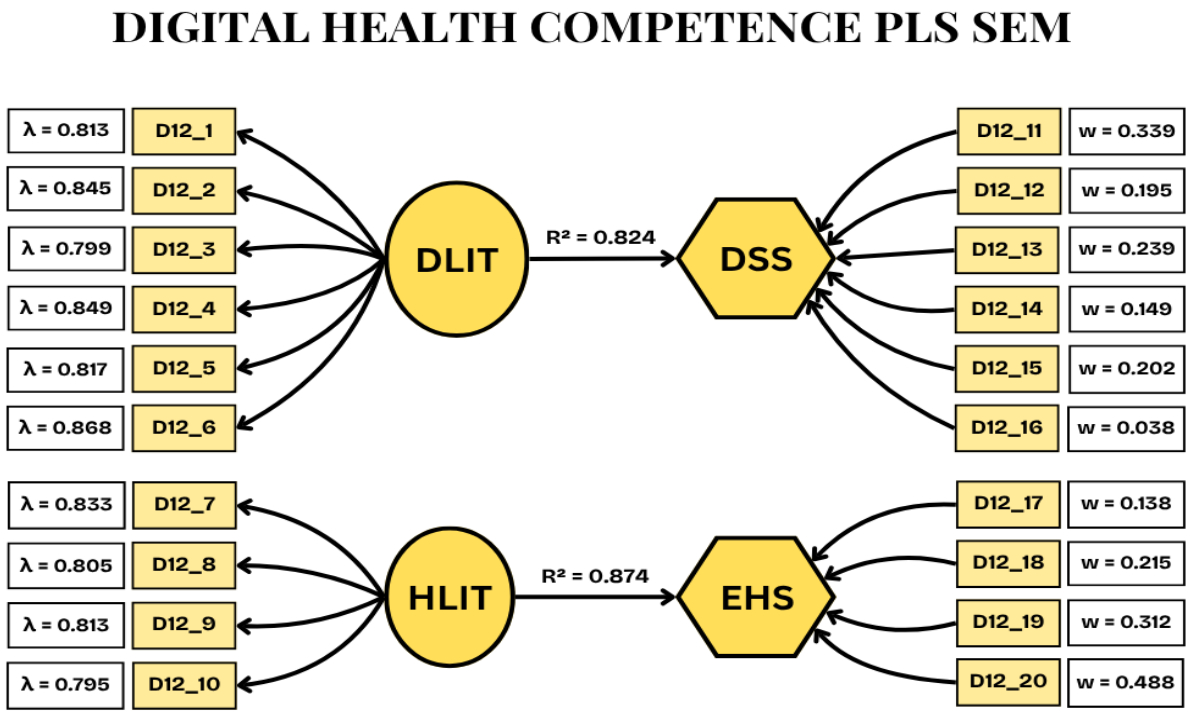

Figure 1: Partial Least Square Structural Equation Modelling of the Digital Health Competence Framework

The framework treated digital health competence not as a purely individual trait, but as a competence emerging from interaction between citizens, healthcare organizations, and the wider health system, in line with recent findings from the field. The results of the PLS SEM, gave us four latent behavioural constructs, namely, Digital Literacy (DLIT), Health Literacy (HLIT), Digital Soft Skills (DSS) and eHealth Skills. Furthermore, the PLS SEM also statistically validated the Digital Competence Framework, as operationalised though the questionnaire. The measurement model showed excellent internal consistency, with Cronbach's alpha values between 0.871 and 0.931 and composite reliability (ρC) above 0.88 for all four constructs. The Average Variance Extracted (AVE)(table 1) values ranged from 0.659 to 0.706, clearly exceeding the recommended 0.50 threshold, thus proving Convergent Validity of the framework. Digital Literacy (DLIT) explained a very large proportion of the variance in Digital Soft Skills (DSS; $R^2 = 0.824$), and Health Literacy (HLIT) accounted for a similarly high share of variance in eHealth Skills (EHS; $R^2 = 0.874$)(figure 1). In both cases, the corresponding path coefficients were very strong (β of 0.908 and 0.935), with t-values above 117, narrow 95% confidence intervals, and p-values below 0.001, indicating highly robust relationships. Considering the overall high reliability, solid convergent validity, acceptable discriminant properties, and theoretically coherent structural paths, it suggested that this novel digital health competence framework can be regarded as statistically validated tool for capturing digital and eHealth competences in this context. Subsequently the Multiple Regression with Restricted Cubic Spline answered our second research question by highlighting the underlying socio demographic factors influencing the latent behavioural constructs. Across both the dependent variable DLIT and HLIT it was found that age influences both in a non-linear way. Compared to the reference baseline, people in Lazio, Puglia, Basilicata and Calabria, on average, have higher digital/health literacy scores, even after adjusting for age, education, profession and other factors. These territorial differences are not trivial compared with the residual spread and suggest that regional infrastructure, service availability, or local initiatives might play a role. Two professions, particularly being a housewife and, in certain models, a general employee with lower skills, are linked to diminished scores. This domain-specific pattern suggests that professional variability affects the gain of several digital health competence. These digital soft skills (DSS), general literacy (DLIT), health-specific literacy (HLIT), and eHealth skills (EHS) may not progress uniformly across different jobs. Pre-Existing Health Condition is a dependable positive correlate exhibiting a strong association with all outcomes, with p-values that are continuously significant. Several of the previous studies have agreed on this insight as well. The results collectively present a cohesive narrative that Digital and eHealth competences differ across three significant dimensions: lifetime (age, in a non-linear fashion), territorial context (region), and occupational setting, with education and pre-existing medical condition offering supplementary, overarching indicators. Conversely, sex, employment status, and amplitude (population size) seem to be insignificant when considering these other variables. The answer to the third research question are provided in detail in the next section.

5.1 Policy Recommendations

The Digital Competence Framework was designed first and foremost as a practical tool for policymakers, enabling them to identify which areas of digital and eHealth competence require immediate attention. In line with much of the existing literature, the present study adopted a predominantly top-down perspective, analysing digital competences at the level of broad populations (e.g., the Italian population as a whole) and large territorial units (e.g., European Union). This approach is useful for mapping overall gaps and inequalities, but our findings suggest that it should now be complemented with more bottom-up strategies. A bottom-up approach would involve focusing on smaller territorial and social units like specific regions, local

health authorities, communities, or occupational groups and examining in detail how digital health interventions are adopted and used in these contexts. Our results confirmed that people in lower socio-economic positions and certain professional categories (housewives, general workers with lower skills) tend to have lower digital and eHealth competence and may therefore require more targeted support. A more granular, bottom-up strategy should also consider the incentive structures faced by different stakeholders in the healthcare value chain. The willingness of citizens, professionals, managers, and technology providers to adopt and sustain digital health interventions depends not only on competence, but also on perceived benefits, costs, risks, and workload implications. Furthermore, the role of artificial intelligence in digital health should be examined within these same context-specific boundaries.

6. Conclusion

This novel Digital Health Competence Framework was found to be a robust instrument for monitoring digital health competence and for guiding targeted interventions. The findings suggest that digital and eHealth competences are unevenly distributed and that certain groups are at greater risk of being left behind in the process of digital transformation of healthcare. Policymakers and organizations can use this framework to identify which territories, professional groups and phase of life require priority support, and to design interventions that are more tailored than generalised digitalisation strategies. The research conducted emphasised the potential and importance of the necessity of the presence of a good level of Digital Health Competence among the citizens of Italy. The framework to understand the levels of competence was operationalised using a questionnaire and interviewing the sample population using telephonic interviews. The framework was found to be statistically valid and further employment of statistical methods helped uncover the socio demographic disparities present in the latent characteristics of the population. The thesis also demonstrated the value of combining complementary statistical approaches to gain a deeper understanding of the task at hand. The synchronous use of PLS-SEM for measurement and structural validation together with multiple regression models, including non-linear spline terms, allowed the analysis to move beyond simple descriptive comparisons and single global scores. This mixed strategy made it possible to unpack how different competence dimensions are structured, how strongly they are linked, and how they vary across population subgroups.

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